

Applied Mathematics for Chemists

(화학수학)

Lecture 12

Differential Equations



Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Three possible cases!

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

(Ideal) Harmonic Oscillator

- Hooke's law

$$F = -kx$$

- Newton's second law

$$F = m \frac{d^2x}{dt^2}$$

- Equation of motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Solving Equation of Motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

- Characteristic equation:

$$\lambda^2 + \frac{k}{m} = 0$$

$$\lambda_1 = i\sqrt{k/m}, \quad \lambda_2 = -i\sqrt{k/m}$$

With $\omega = \sqrt{k/m}$, the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

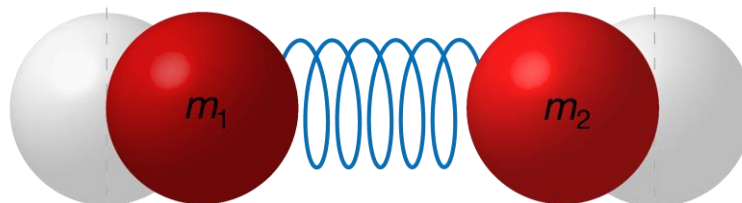
Assume that we know the conditions at $t = 0$:

$$x(0) = 0, \quad v(0) = \left. \frac{dx}{dt} \right|_{t=0} = v_0$$

The solution is

$$x(t) = \frac{v_0}{\omega} \sin \omega t$$

Vibration of Diatomic Molecules



- Reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

- Vibrational frequency

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

(Image: <http://felix.chem.washington.edu/455/vibrations.html>)

Damped Harmonic Oscillator

$$F = m \frac{d^2 x}{dt^2} = -kx - \zeta \frac{dx}{dt}$$

- Standard form

$$\frac{d^2 x}{dt^2} + \frac{\zeta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

- Characteristic equation

$$\lambda^2 + \frac{\zeta}{m} \lambda + \frac{k}{m} = 0$$

(1) $D > 0$: Overdamping (과감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

- Note that $\lambda_1, \lambda_2 < 0$ (Example 12.6) → decay!

(2) $D = 0$: Critical Damping (임계감쇠)

$$\lambda_1 = -\frac{\zeta}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$$

- Note that $\lambda_1 < 0 \rightarrow$ decay!

(3) $D < 0$: Underdamping (미급감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

Let $\omega = \sqrt{(k/m) - (\zeta/2m)^2}$;

$$\lambda_1 = -\frac{\zeta}{2m} + i\omega, \quad \lambda_2 = -\frac{\zeta}{2m} - i\omega$$

$$x(t) = e^{-\zeta t/2m} (c_1 \cos \omega t + c_2 \sin \omega t)$$

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Guess for Particular Solution

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$A e^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t} (A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Today's Class

- Inhomogeneous linear ODEs with constant coefficients
 - Application: forced harmonic oscillator
- Laplace transform and its application in DE
 - Laplace transform
 - Use of Laplace transform in solving DEs
 - Application: underdamped harmonic oscillator

Application: Forced Harmonic Oscillator

$$F = m \frac{d^2 x}{dt^2} = -kx + \underbrace{F(t)}_{\text{external force}}$$

- Standard form

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = \frac{F(t)}{m}$$

- Complementary equation

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

$$\rightarrow x_c(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

The equation of motion is

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = \frac{F(t)}{m} = \frac{F_0}{m} \sin \alpha t$$

The trial function for $\sin \alpha t$ is $A \sin \alpha t + B \cos \alpha t$. Substitution gives

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m} (A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m}(A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Rearranging the equation,

$$\left(-\alpha^2 + \frac{k}{m}\right)A \sin \alpha t + \left(-\alpha^2 + \frac{k}{m}\right)B \cos \alpha t = \frac{F_0}{m} \sin \alpha t$$

Hence, $B = 0$ and $A = \frac{F_0/m}{-\alpha^2 + k/m} = \frac{F_0}{k - m\alpha^2}$.

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

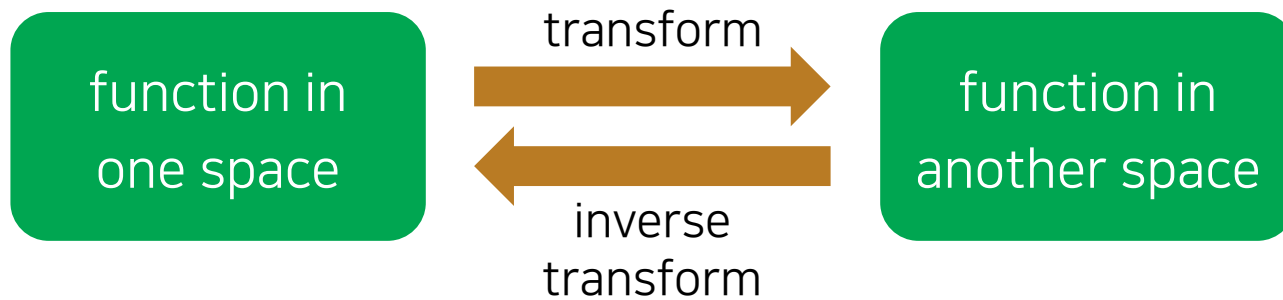
where F_0 and α are constants. Find the general solution to the equation of motion.

$$x_p = \frac{F_0}{k - m\alpha^2} \sin \alpha t$$

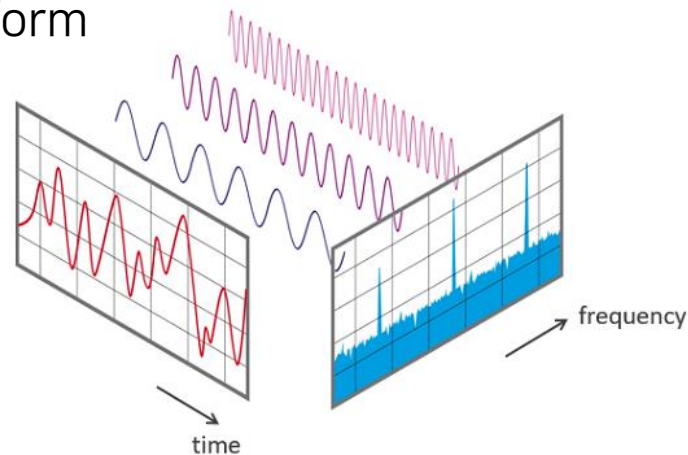
As we know $x_c(t) = c_1 \cos \omega t + c_2 \sin \omega t$, the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{k - m\alpha^2} \sin \alpha t$$

Integral Transforms



e.g. Fourier transform



(Image: <https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft>)

Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Example 11.7

Find the Laplace transform of the function $f(t) = t^2$.

From the definition of the Laplace transform,

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} t^2 e^{-st} dt$$

By setting $u = st$ and using the table of integrals,

$$F(s) = \frac{1}{s^3} \int_0^{\infty} u^2 e^{-u} du = \frac{2!}{s^3} = \frac{2}{s^3}$$

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos(at)$
$\frac{a}{s^2 + a^2}$	$\sin(at)$
$\frac{s}{s^2 - a^2}$	$\cosh(at)$
$\frac{a}{s^2 - a^2}$	$\sinh(at)$

Shifting Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

Example 11.8

Use the shifting theorem to obtain the Laplace transform of the function

$$f(t) = e^{at} \cos(kt).$$

From Table 11.1,

$$\mathcal{L}\{\cos(kt)\} = \frac{s}{s^2 + k^2}$$

Hence, from the shifting theorem, $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$,

$$\mathcal{L}\{e^{at} \cos(kt)\} = \frac{s - a}{(s - a)^2 + k^2}$$

Derivative Theorem and Integral Theorem

- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$

Example 11.9

Find the inverse Laplace transform of

$$F(s) = \frac{1}{s(s-a)}.$$

From Table 11.1,

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

The integral theorem states that

$$\mathcal{L}\left\{\int_0^t f(u) \, du\right\} = \frac{1}{s} \mathcal{L}\{f(t)\}$$

Hence,

$$F(s) = \frac{1}{s(s-a)} = \frac{1}{s} \mathcal{L}\{e^{at}\} = \mathcal{L}\left\{\int_0^t e^{au} \, du\right\}$$

Example 11.9

Find the inverse Laplace transform of

$$F(s) = \frac{1}{s(s-a)}.$$

$$F(s) = \mathcal{L} \left\{ \int_0^t e^{au} \, du \right\}$$

The inverse gives

$$\mathcal{L}^{-1}\{F(s)\} = \int_0^t e^{au} \, du = \frac{1}{a} e^{au} \Big|_0^t = \frac{1}{a} (e^{at} - 1)$$

Use of Laplace Transforms in DE

- (Generalized) derivative theorem

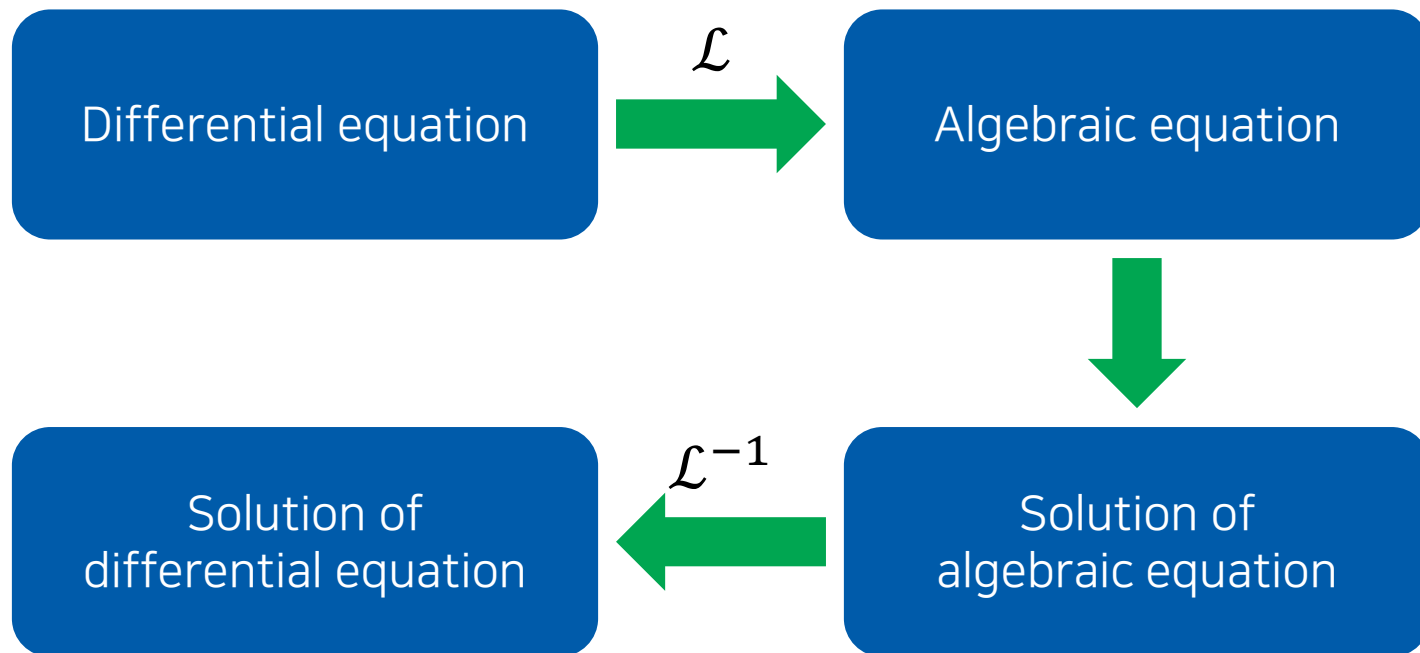
$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

$$\mathcal{L}\left\{\frac{d^2f(t)}{dt^2}\right\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$$

differential equation $\rightarrow$ algebraic equation

Solving DE with Laplace Transforms



Example

Use the Laplace transform to solve the initial-value problem

$$\frac{dx}{dt} + 3x = 13 \sin 2t, \quad x(0) = 6.$$

Take the Laplace transform on each term:

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} + 3\mathcal{L}\{x\} = 13\mathcal{L}\{\sin 2t\}$$

Use the known transforms:

$$[s\mathcal{L}\{x\} - x(0)] + 3\mathcal{L}\{x\} = \frac{26}{s^2 + 4}$$

$$(s + 3)\mathcal{L}\{x\} = \frac{26}{s^2 + 4} + 6$$

Example

Use the Laplace transform to solve the initial-value problem

$$\frac{dx}{dt} + 3x = 13 \sin 2t, \quad x(0) = 6.$$

$$(s + 3)\mathcal{L}\{x\} = \frac{26}{s^2 + 4} + 6$$

Thus,

$$\mathcal{L}\{x\} = \frac{26}{(s + 3)(s^2 + 4)} + \frac{6}{s + 3}$$

$$\mathcal{L}\{x\} = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)}$$

Example

Use the Laplace transform to solve the initial-value problem

$$\frac{dx}{dt} + 3x = 13 \sin 2t, \quad x(0) = 6.$$

$$\mathcal{L}\{x\} = \frac{6s^2 + 50}{(s + 3)(s^2 + 4)}$$

Applying the method of partial fractions,

$$\frac{6s^2 + 50}{(s + 3)(s^2 + 4)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 4} = \frac{A(s^2 + 4) + (Bs + C)(s + 3)}{(s + 3)(s^2 + 4)}$$

As $A = 8$, $B = -2$, $C = 6$,

$$\mathcal{L}\{x\} = \frac{8}{s + 3} + \frac{-2s}{s^2 + 4} + \frac{6}{s^2 + 4}$$

Example

Use the Laplace transform to solve the initial-value problem

$$\frac{dx}{dt} + 3x = 13 \sin 2t, \quad x(0) = 6.$$

$$\mathcal{L}\{x\} = \frac{8}{s+3} + \frac{-2s}{s^2+4} + \frac{6}{s^2+4}$$

Hence,

$$\mathcal{L}\{x\} = 8\mathcal{L}\{e^{-3t}\} - 2\mathcal{L}\{\cos 2t\} + 3\mathcal{L}\{\sin 2t\}$$

and

$$x(t) = 8e^{-3t} - 2 \cos 2t + 3 \sin 2t$$

Application: Underdamped Oscillator

$$\frac{d^2x}{dt^2} + \frac{\zeta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

Apply the Laplace transform:

$$[s^2 \mathcal{L}\{x\} - sx(0) - x'(0)] + \frac{\zeta}{m} [s \mathcal{L}\{x\} - x(0)] + \frac{k}{m} \mathcal{L}\{x\} = 0$$

Solving the algebraic equation,

$$\mathcal{L}\{x\} = \frac{sx(0) + x'(0) + (\zeta/m)x(0)}{s^2 + (\zeta/m)s + (k/m)}$$

Rearrangement

$$\begin{aligned}\mathcal{L}\{x\} &= \frac{sx(0) + x'(0) + (\zeta/m)x(0)}{s^2 + (\zeta/m)s + (k/m)} \\ &= \frac{x(0)(s + \zeta/m) + x'(0)}{(s + \zeta/2m)^2 - \zeta^2/4m^2 + k/m}\end{aligned}$$

Let $a = \zeta/2m$ and $\omega^2 = k/m - \zeta^2/4m^2 > 0$ (underdamping).

$$\begin{aligned}\mathcal{L}\{x\} &= \frac{x(0)(s + 2a) + x'(0)}{(s + a)^2 + \omega^2} \\ &= \frac{x(0)(s + a)}{(s + a)^2 + \omega^2} + \frac{ax(0) + x'(0)}{(s + a)^2 + \omega^2}\end{aligned}$$

Known Facts

$$\mathcal{L}\{x\} = \frac{x(0)(s + a)}{(s + a)^2 + \omega^2} + \frac{ax(0) + x'(0)}{(s + a)^2 + \omega^2}$$

- Shifting theorem

$$\mathcal{L}\{e^{-at}f(t)\} = F(s + a)$$

- Known transforms

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}\{x\} = x(0)\mathcal{L}\{e^{-at} \cos \omega t\} + \frac{ax(0) + x'(0)}{\omega} \mathcal{L}\{e^{-at} \sin \omega t\}$$

Final Solution

$$\mathcal{L}\{x\} = x(0)\mathcal{L}\{e^{-at} \cos \omega t\} + \frac{ax(0) + x'(0)}{\omega} \mathcal{L}\{e^{-at} \sin \omega t\}$$

Hence, the inverse transform gives

$$x(t) = x(0)e^{-at} \cos \omega t + \frac{ax(0) + x'(0)}{\omega} e^{-at} \sin \omega t$$

$$x(t) = e^{-\zeta t/2m} \left[\underbrace{x(0)}_{c_1} \cos \omega t + \underbrace{\frac{ax(0) + x'(0)}{\omega}}_{c_2} \sin \omega t \right]$$